Ch 10.1: Neural Nets

Lecture 29 - CMSE 381

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Mon, Nov 18, 2024

Announcements

Last time:

- SVM

This lecture:

- Feed Forward Neural Nets

Announcements:

- Homework #9 due Sunday, Nov 24
- No class wednesday Nov 27. Virtual office hours by request.

Lec #	Date			Reading	HW
21	Mon	10/28	Polynomial & Step Functions	7.1,7.2	
22	Wed	10/30	Step Functions; Basis functions; Start Splines	7.2 - 7.4	
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27	Mon	11/11	Maximal Margin Classifier	9.1	
28	Wed	11/13	SVC	9.2	
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33	Mon	11/25	TBD: Unsupervised learning/clustering	12.1, 12.4?	
	Wed	11/27	Virtual: Project office hours		
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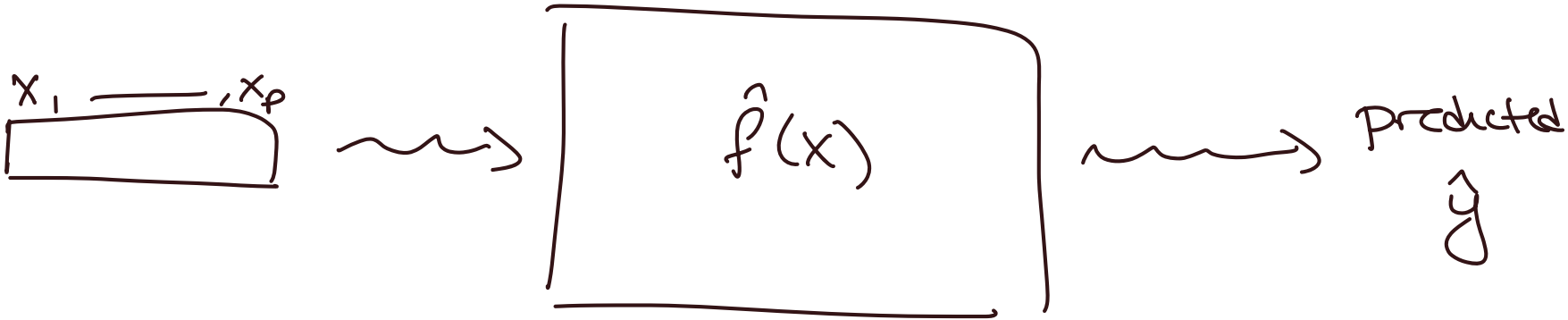
	Mon	12/2	Review		
	Wed	12/4	**Midterm #3**		
	Fri	12/6	No class - EGR Design Day		Project due

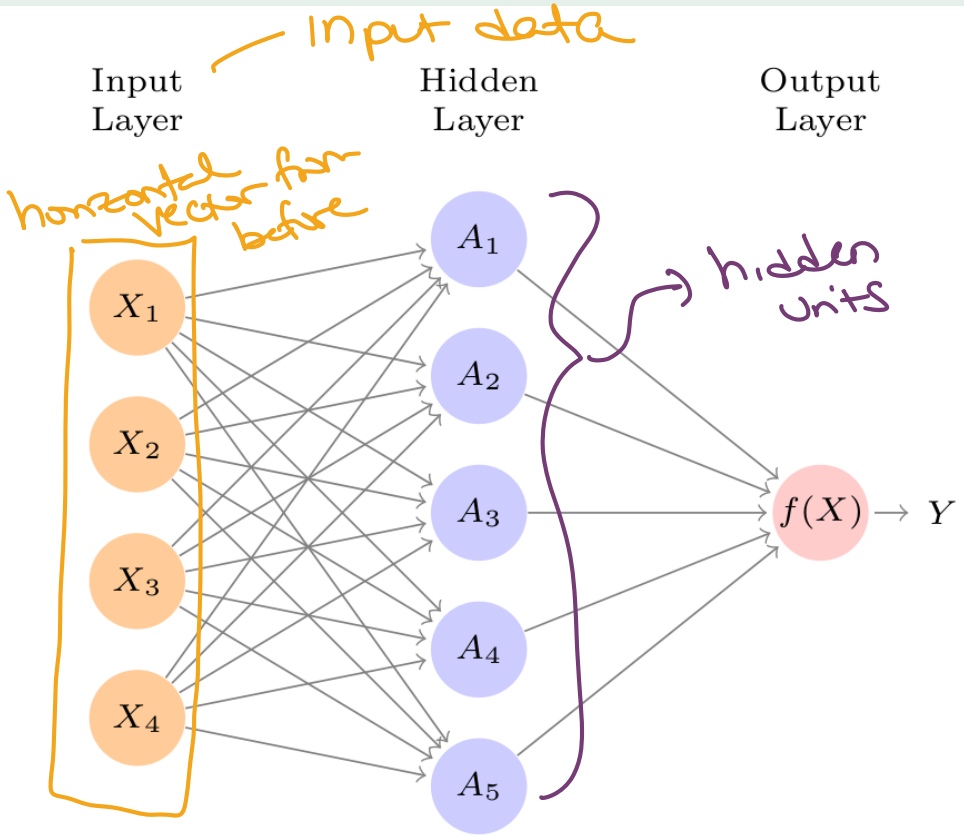
Section 1

Neural Nets

The idea



Feed Forward Neural Network: The cartoon



Arrows mean:

Take linear combo of X_i's to get each A_i

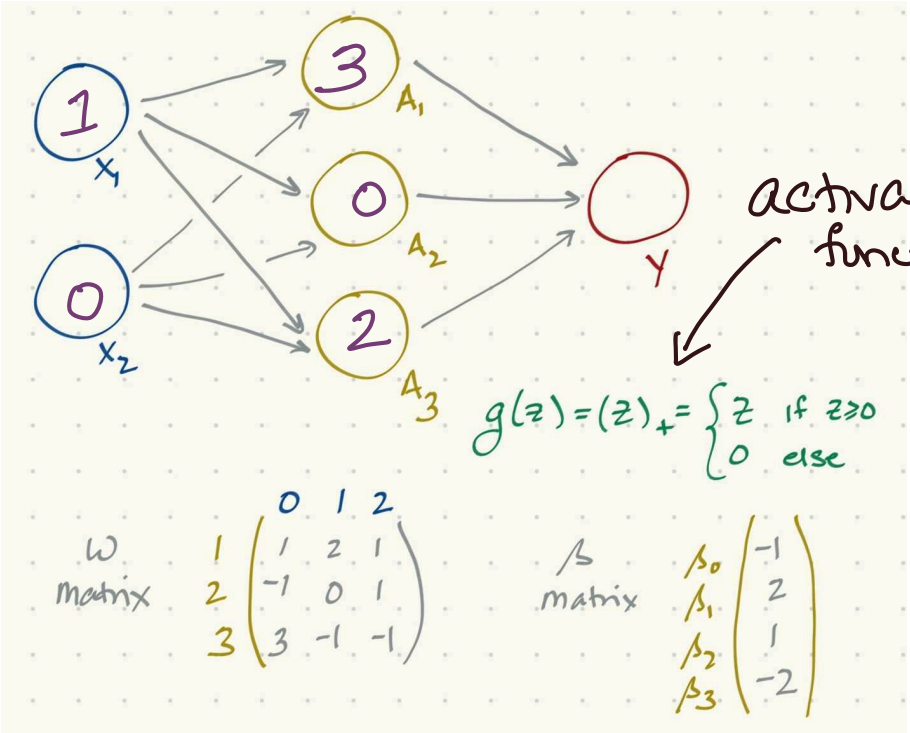
Today: one hidden layer

We pick how many "hidden units" we have

A very simple example

Computing A_k for $(1, 0)$

$$A_k = h_k(X) = g(w_{k0} + \sum_{j=1}^{p} w_{kj} X_j),$$



$$\begin{aligned} A_1 &= g(w_{10} + w_{11} X_1 + w_{12} X_2) \\ &= g(1 + 2X_1 + 1X_2) \\ &= g(1 + 2(1) + 1(0)) \\ &= g(1 + 2) = g(3) = 3 \end{aligned}$$

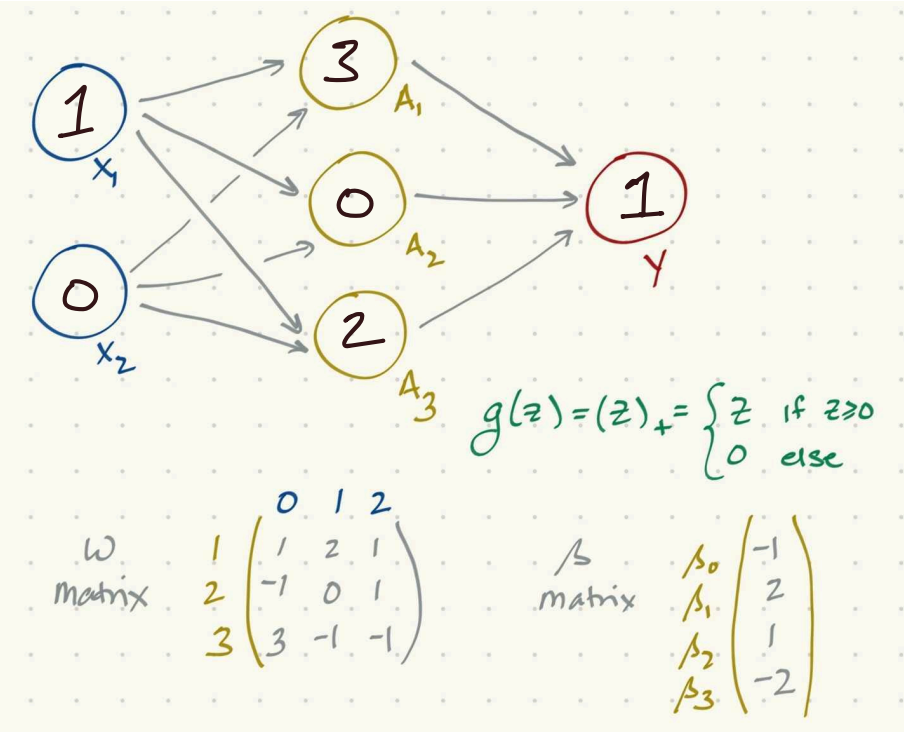
$$\begin{aligned} A_2 &= g(w_{20} + w_{21} X_1 + w_{22} X_2) \\ &= g(-1 + 0(1) + 1(0)) \\ &= g(-1) = 0 \end{aligned}$$

$$\begin{aligned} A_3 &= g(w_{30} + w_{31} X_1 + w_{32} X_2) \\ &= g(3 + (-1)(1) + (-1)(0)) \\ &= g(2) = 2 \end{aligned}$$

A very simple example

Computing Y for $(1, 0)$

$$f(X) = \beta_0 + \sum_{k=1}^{K} \beta_k A_k$$



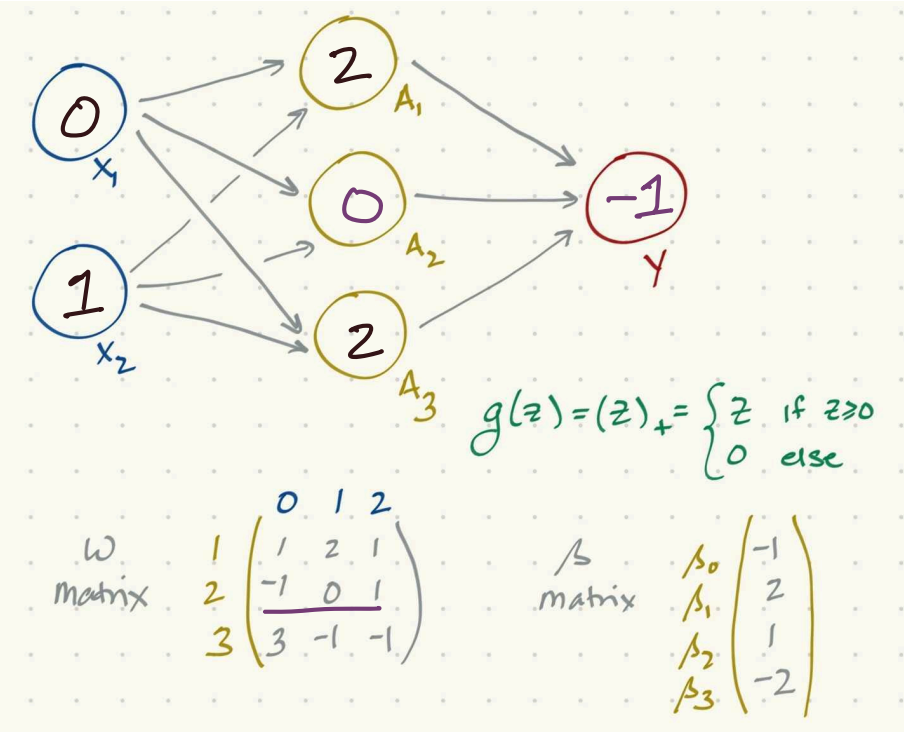
$$f(x) = \beta_0 + \beta_1 A_1 + \beta_2 A_2 + \beta_3 A_3$$

$$f(1,0) = -1 + 2(3) + 1(0) + (-2)(2)$$
$$= -1 + 6 - 4 = 1$$

A very simple example

Computing Y for $(0, 1)$

$$A_k = h_k(X) = g(w_{k0} + \textstyle\sum_{j=1}^{p} w_{kj} X_j), \qquad f(X) = \beta_0 + \sum_{k=1}^{K} \beta_k A_k$$



$$A_2 = g(w_{20} + w_{21} X_1 + w_{22} X_2)$$
$$= g(-1 + 0 \cdot 0 + 1 \cdot 1)$$
$$= g(0) = 0$$

$$Y = \beta_0 + \beta_1 A_1 + \beta_2 A_2 + \beta_3 A_3$$
$$= -1 + 2 \cdot 2 + 1 \cdot 0 + (-2) \cdot 2$$
$$= -1 + 4 + 0 - 4 = -1$$

A different example

- Draw the diagram for a neural net with input data points with $p = 3$ (i.e., (X_1, X_2, X_3)) and two units in the hidden layer.
- Using the ω and β matrices, what is the output predicted Y for the point $(2, 0, 1)$?

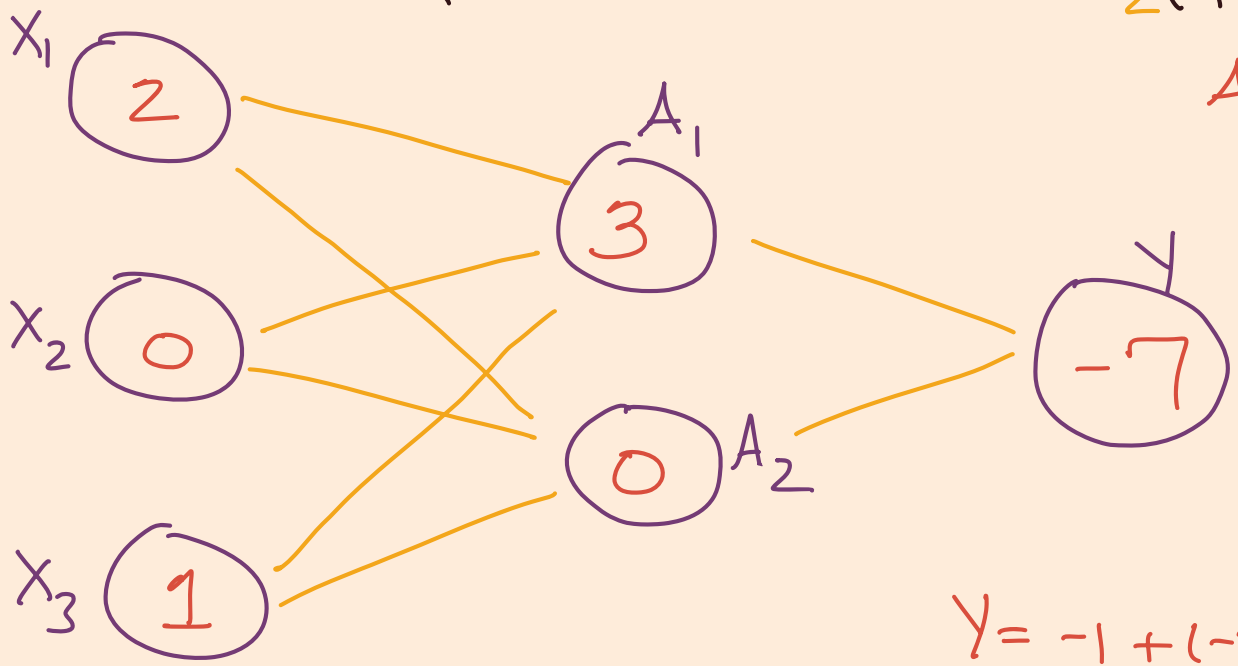
$$\omega = \begin{pmatrix} 1 & 0 & -2 & 2 \\ -3 & 1 & 0 & -1 \end{pmatrix} \qquad \beta = \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}$$

- Use the activation function

$$g(z) = (z)_+ = \begin{cases} 0 & \text{if } z < 0 \\ z & \text{else.} \end{cases}$$

Extra space

$$w = \begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 1 & 1 & 0 & -2 & 2 \\ 2 & -3 & 1 & 0 & -1 \end{array} \qquad s = \begin{array}{c|c} 0 & -1 \\ 1 & -2 \\ 2 & 1 \end{array}$$

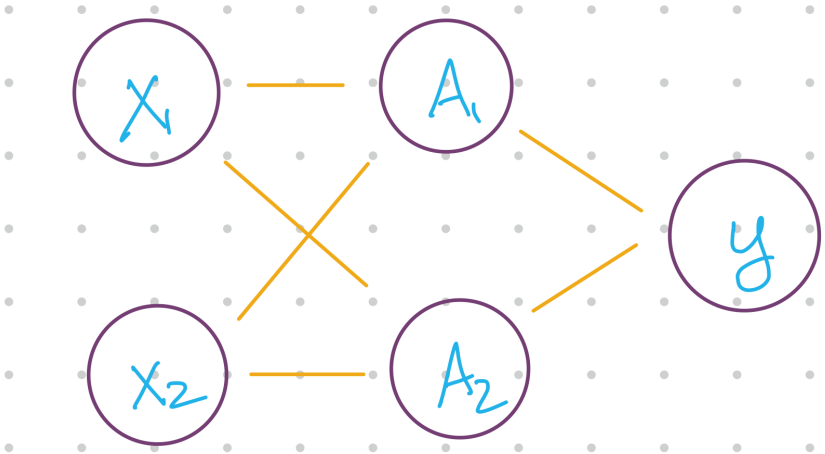


$$A_1 = g(1 + 2(0) + (-2)0 + 1(2))$$
$$= g(3) = 3$$

$$A_2 = g(-3 + 2(1) + 0(0) + (-1)(1))$$
$$= g(-3 + 2 - 1)$$
$$= g(-2) = 0$$

$$Y = -1 + (-2)(3) + (1)(0)$$
$$= -1 - 6 + 0 = -7$$

What if there's no activation function?



$$\omega = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \qquad \beta = \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}$$

ex) $A_1 = 1 + 2x_1 + 3x_2$

$A_2 = 4 + 5x_1 + 6x_2$

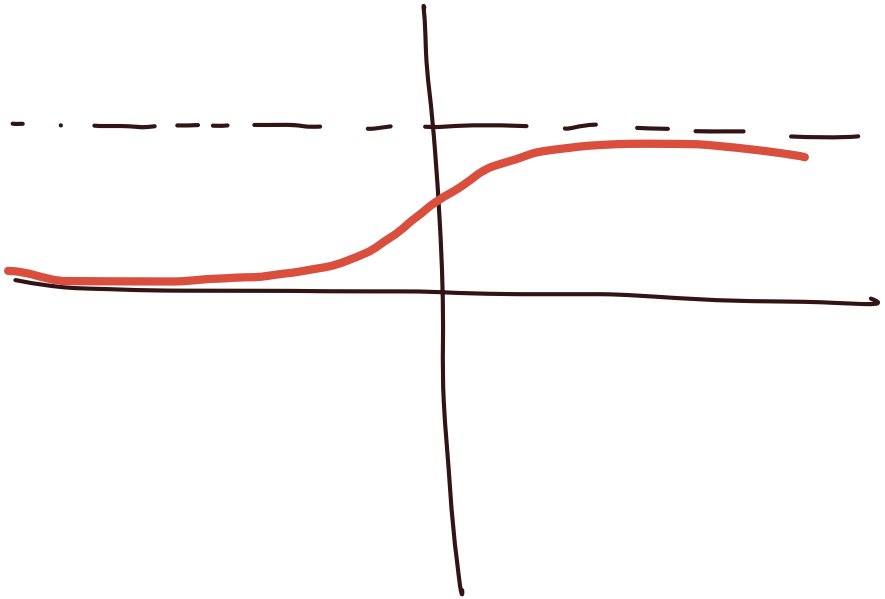
$$\begin{aligned} Y &= -1 + (-2)A_1 + 1(A_2) \\ &= -1 - 2 - 4x_1 - 6x_2 \\ &\quad + 4 + 5x_1 + 6x_2 \\ &= 1 + 1x_1 + 0x_2 \end{aligned}$$

This is just a linear function of (x_1, x_2), so we could have done this day 3

Choices for activation function

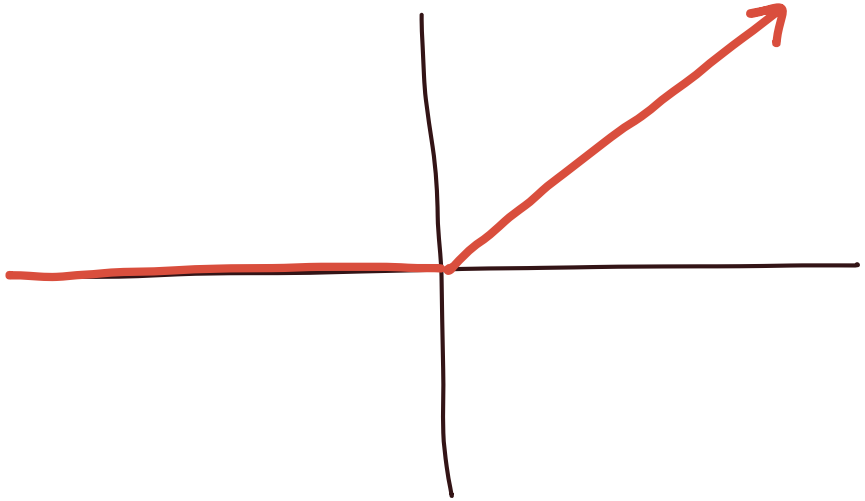
Sigmoid:

$$g(z) = \frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}}$$

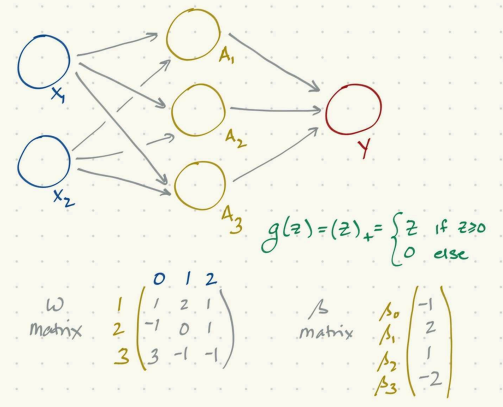


ReLU: Rectified linear unit

$$g(z) = (z)_+ = \begin{cases} 0 & \text{if } z < 0 \\ z & \text{else.} \end{cases}$$



Matrix version: First layer

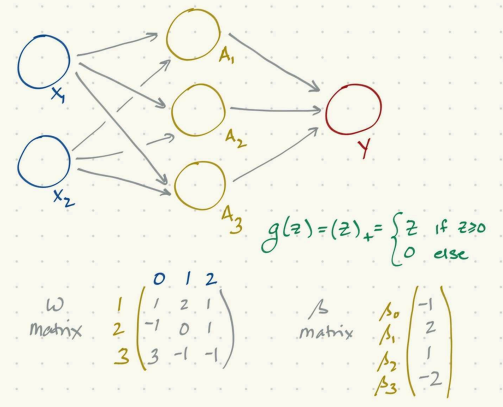


$$A_k = h_k(X) = g(w_{k0} + \sum_{j=1}^{p} w_{kj} X_j),$$

$$A = g(\mathbf{W} \cdot \mathbf{X}) \qquad \mathbf{X}^T = (1 \; X_1 \; X_2 \; \cdots \; X_p)$$

$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{pmatrix} \quad & \quad & \quad \\ & W & \\ \quad & \quad & \quad \end{pmatrix} \end{array} \begin{pmatrix} 1 \\ X_1 \\ X_2 \\ X \end{pmatrix}$$

Matrix version: Output



$$f(X) = \beta_0 + \sum_{k=1}^{K} \beta_k A_k$$

$$Y = \beta \cdot \mathbf{A} \qquad \mathbf{A}^T = (1 \; A_1 \; A_2 \; \cdots \; A_K)$$

$$(\beta_0 \; \beta_1 \; \beta_2 \; \beta_3) \begin{pmatrix} 1 \\ A_1 \\ A_2 \\ A_3 \end{pmatrix}$$

Now what?

Choose parameters by minimizing RSS, $\sum_{i=1}^{n}(y_i - f(x_i))^2$

Chosen in advance:

- Number of hidden layers
- Number of hidden units in each layer
- Activation function g

"Architecture"

Tuned by the model:

W

β

Coding

Next time

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